Linux Kernel TCP/IP Stack & Routing Implementation

Coverage: Linux Network Packet Flow, sk_buff Structure, net_device, Forwarding Information Base (FIB), Route Cache, iproute2

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1. Linux Kernel Networking

- sk_buff: Socket buffer with head, data, tail, end pointers for zero-copy header manipulation.
- FIB: Forwarding Information Base implemented as a high-speed Level-Compressed Trie (LC-trie).

2. Hands-on iproute2

- ip route add, ip rule add for Policy-Based Routing (PBR).

■ KTU University High-Yield Exam Questions

- Q: Explain the architecture and role of sk_buff in Linux kernel packet processing. (8 Marks)